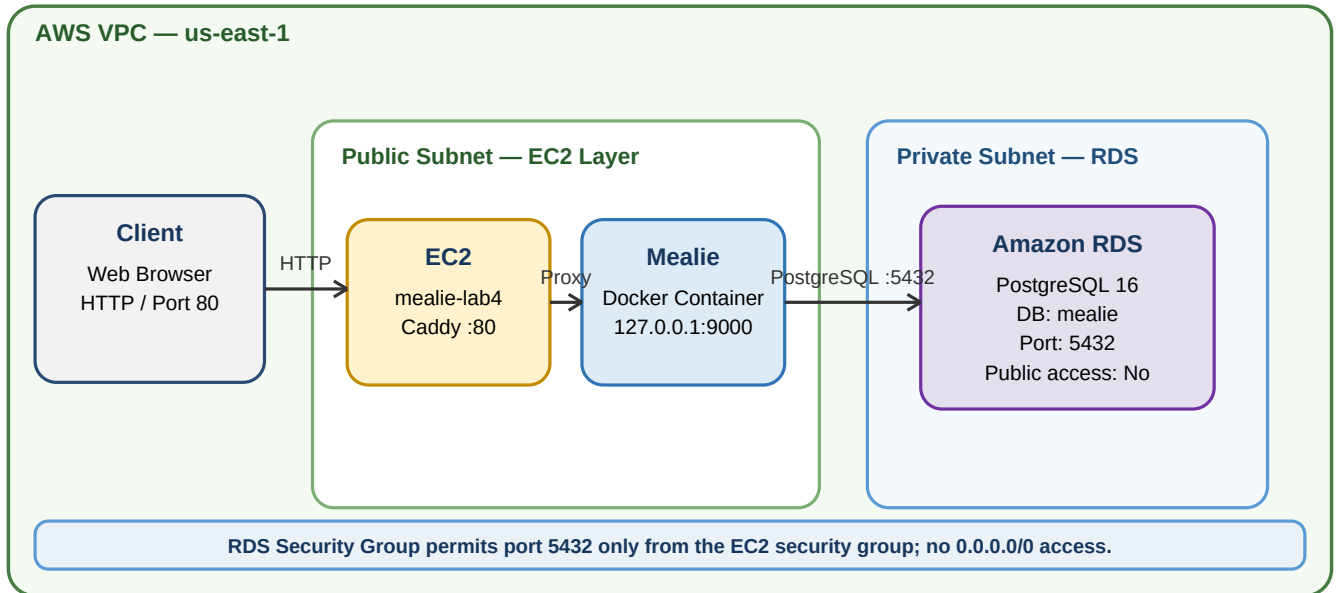


Lab 5 — Assignment 1: RDS PostgreSQL Deployment

Deploy and Connect Database for the Lab 4 EC2 Application (Mealie Recipe Manager)

1. Architecture

The Lab 4 Mealie application continues to run on an Amazon EC2 instance using a Docker container and Caddy reverse proxy. The PostgreSQL database was migrated from the EC2 instance to a managed Amazon RDS PostgreSQL database in the same VPC. The client request path is: **Client** → **Caddy on EC2** → **Mealie container** → **Amazon RDS PostgreSQL**.



2. Database Settings

Setting	Value
Engine	PostgreSQL 16
Instance identifier	mealie-db
Instance class	db.t3.micro (Free Tier)
Storage	20 GiB
Public accessibility	No
Database name	mealie
Endpoint	mealie-db.c8b80go427hx.us-east-1.rds.amazonaws.com
Port	5432

3. Security Settings

A dedicated security group named **mealie-rds-sg** was created for the RDS instance. Inbound PostgreSQL access is restricted to the Lab 4 EC2 instance's security group. The RDS instance is configured as not publicly accessible, providing an additional layer of protection.

Type	Port	Source
PostgreSQL	5432	EC2 security group sg-010134994e59ab4eb only

4. Data Migration

The existing Lab 4 database was migrated to RDS using **pg_dump** and **psql**. The migration preserved the existing data rather than starting with an empty database.

Database dump:

```
pg_dump -U mealie -h 127.0.0.1 mealie > mealie-migrate.sql
```

Restore into RDS:

```
psql -h mealie-db.c8b80go427hx.us-east-1.rds.amazonaws.com -U mealie -d mealie -f mealie-migrate.sql
```

Post-migration verification confirmed that the data was transferred correctly: **3 recipes and 1 user** were present, matching the row counts from the Lab 4 local database.

5. Application Reconnection

Mealie's environment configuration was updated to point to the RDS endpoint. The relevant configuration was:

```
POSTGRES_SERVER=mealie-db.c8b80go427hx.us-east-1.rds.amazonaws.com
POSTGRES_PORT=5432 POSTGRES_DB=mealie POSTGRES_USER=mealie
```

The container was restarted and the application was verified using:

```
sudo docker restart mealie
curl -s http://127.0.0.1:9000/api/app/about
```

The response confirmed that the Mealie application was running successfully after connecting to RDS.

6. CRUD Demonstration

Operation	Action performed
Create	Added a new test recipe through the Mealie UI
Read	Opened an existing migrated recipe, such as Chicken Parmesan
Update	Edited the test recipe's title and ingredients
Delete	Removed the test recipe

All four operations were performed through the running EC2 application and confirmed that the application could read from and write to the RDS-backed database.

7. Evidence Summary

- EC2 application running and reachable through its public IP.
- RDS instance showing an Available status.
- RDS configured with public accessibility disabled.
- Security group allowing PostgreSQL access only from the EC2 security group.
- Successful **pg_dump** and restore output with row-count verification.
- Mealie application successfully connected to the RDS endpoint.
- Create, Read, Update, and Delete operations performed live in the browser.

Conclusion

The Lab 4 Mealie application was successfully connected to Amazon RDS PostgreSQL. The local database was migrated, access was secured through security groups, and CRUD operations were verified through the application. This demonstrates the use of a managed cloud database with an EC2-hosted application.